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On general relativity:

python computations.

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Contents

Page

Contents **ii**

1 Exterior Schwarschild metric **3**
1.1 Christoffel symbols 3
1.2 Riemann tensor 6
1.3 Ricci tensor 15
1.4 Ricci scalar 16
1.5 Einstein tensor 16

2 Interior Schwarschild metric **17**
2.1 Christoffel symbols 17
2.2 Riemann tensor 20
2.3 Ricci tensor 29
2.4 Ricci scalar 29
2.5 Einstein tensor 30

3 FRW metric **31**
3.1 Christoffel symbols 31
3.2 Riemann tensor 34
3.3 Ricci tensor 43
3.4 Ricci scalar 44
3.5 Einstein tensor 44

Abstract

In these notes, we will compute general relativity with the help of python.

Chapter 1

Exterior Schwarschild metric

The metric is

$$g = \begin{bmatrix} -A(r) & 0 & 0 & 0 \\ 0 & B(r) & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix} ,$$

.

1.1 Christoffel symbols

The Christoffel symbols are

$$\Gamma^0_{00} = 0 ,$$

$$\Gamma^0_{01} = \frac{0.5 \frac{d}{dr} A(r)}{A(r)} ,$$

$$\Gamma^0_{02} = 0 ,$$

$$\Gamma^0_{03} = 0 ,$$

$$\Gamma^0_{10} = \frac{0.5 \frac{d}{dr} A(r)}{A(r)} ,$$

$$\Gamma^0_{11} = 0 ,$$

$$\Gamma^0_{12} = 0 ,$$

$$\Gamma^0_{13} = 0 ,$$

$$\Gamma^0_{20} = 0 ,$$

$$\Gamma^0_{21} = 0 ,$$

$$\Gamma^0_{22} = 0 ,$$

$$\Gamma^0_{23} = 0 ,$$

$$\Gamma^0_{30} = 0 \ ,$$
$$\Gamma^0_{31} = 0 \ ,$$
$$\Gamma^0_{32} = 0 \ ,$$
$$\Gamma^0_{33} = 0 \ ,$$
$$\Gamma^1_{00} = \frac{0.5\frac{d}{dr}A(r)}{B(r)} \ ,$$
$$\Gamma^1_{01} = 0 \ ,$$
$$\Gamma^1_{02} = 0 \ ,$$
$$\Gamma^1_{03} = 0 \ ,$$
$$\Gamma^1_{10} = 0 \ ,$$
$$\Gamma^1_{11} = \frac{0.5\frac{d}{dr}B(r)}{B(r)} \ ,$$
$$\Gamma^1_{12} = 0 \ ,$$
$$\Gamma^1_{13} = 0 \ ,$$
$$\Gamma^1_{20} = 0 \ ,$$
$$\Gamma^1_{21} = 0 \ ,$$
$$\Gamma^1_{22} = -\frac{1.0r}{B(r)} \ ,$$
$$\Gamma^1_{23} = 0 \ ,$$
$$\Gamma^1_{30} = 0 \ ,$$
$$\Gamma^1_{31} = 0 \ ,$$
$$\Gamma^1_{32} = 0 \ ,$$
$$\Gamma^1_{33} = -\frac{1.0r\sin^2(\theta)}{B(r)} \ ,$$
$$\Gamma^2_{00} = 0 \ ,$$
$$\Gamma^2_{01} = 0 \ ,$$
$$\Gamma^2_{02} = 0 \ ,$$
$$\Gamma^2_{03} = 0 \ ,$$
$$\Gamma^2_{10} = 0 \ ,$$
$$\Gamma^2_{11} = 0 \ ,$$
$$\Gamma^2_{12} = \frac{1.0}{r} \ ,$$

$$\Gamma^2_{13} = 0 \ ,$$
$$\Gamma^2_{20} = 0 \ ,$$
$$\Gamma^2_{21} = \frac{1.0}{r} \ ,$$
$$\Gamma^2_{22} = 0 \ ,$$
$$\Gamma^2_{23} = 0 \ ,$$
$$\Gamma^2_{30} = 0 \ ,$$
$$\Gamma^2_{31} = 0 \ ,$$
$$\Gamma^2_{32} = 0 \ ,$$
$$\Gamma^2_{33} = -0.5 \sin(2\theta) \ ,$$
$$\Gamma^3_{00} = 0 \ ,$$
$$\Gamma^3_{01} = 0 \ ,$$
$$\Gamma^3_{02} = 0 \ ,$$
$$\Gamma^3_{03} = 0 \ ,$$
$$\Gamma^3_{10} = 0 \ ,$$
$$\Gamma^3_{11} = 0 \ ,$$
$$\Gamma^3_{12} = 0 \ ,$$
$$\Gamma^3_{13} = \frac{1.0}{r} \ ,$$
$$\Gamma^3_{20} = 0 \ ,$$
$$\Gamma^3_{21} = 0 \ ,$$
$$\Gamma^3_{22} = 0 \ ,$$
$$\Gamma^3_{23} = \frac{1.0}{\tan(\theta)} \ ,$$
$$\Gamma^3_{30} = 0 \ ,$$
$$\Gamma^3_{31} = \frac{1.0}{r} \ ,$$
$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)} \ ,$$
$$\Gamma^3_{33} = 0 \ ,$$
$$.$$

1.2 Riemann tensor

The Riemann tensor is

$$R^0_{000} = 0 \ ,$$
$$R^0_{001} = 0 \ ,$$
$$R^0_{002} = 0 \ ,$$
$$R^0_{003} = 0 \ ,$$
$$R^0_{010} = 0 \ ,$$
$$R^0_{011} = 0 \ ,$$
$$R^0_{012} = 0 \ ,$$
$$R^0_{013} = 0 \ ,$$
$$R^0_{020} = 0 \ ,$$
$$R^0_{021} = 0 \ ,$$
$$R^0_{022} = 0 \ ,$$
$$R^0_{023} = 0 \ ,$$
$$R^0_{030} = 0 \ ,$$
$$R^0_{031} = 0 \ ,$$
$$R^0_{032} = 0 \ ,$$
$$R^0_{033} = 0 \ ,$$
$$R^0_{100} = 0 \ ,$$
$$R^0_{101} = -\frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)} \ ,$$
$$R^0_{102} = 0 \ ,$$
$$R^0_{103} = 0 \ ,$$
$$R^0_{110} = \frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)} \ ,$$
$$R^0_{111} = 0 \ ,$$
$$R^0_{112} = 0 \ ,$$
$$R^0_{113} = 0 \ ,$$
$$R^0_{120} = 0 \ ,$$
$$R^0_{121} = 0 \ ,$$

$$R^0_{122} = 0 \ ,$$
$$R^0_{123} = 0 \ ,$$
$$R^0_{130} = 0 \ ,$$
$$R^0_{131} = 0 \ ,$$
$$R^0_{132} = 0 \ ,$$
$$R^0_{133} = 0 \ ,$$
$$R^0_{200} = 0 \ ,$$
$$R^0_{201} = 0 \ ,$$
$$R^0_{202} = -\frac{0.5r\frac{d}{dr}A(r)}{A(r)B(r)} \ ,$$
$$R^0_{203} = 0 \ ,$$
$$R^0_{210} = 0 \ ,$$
$$R^0_{211} = 0 \ ,$$
$$R^0_{212} = 0 \ ,$$
$$R^0_{213} = 0 \ ,$$
$$R^0_{220} = \frac{0.5r\frac{d}{dr}A(r)}{A(r)B(r)} \ ,$$
$$R^0_{221} = 0 \ ,$$
$$R^0_{222} = 0 \ ,$$
$$R^0_{223} = 0 \ ,$$
$$R^0_{230} = 0 \ ,$$
$$R^0_{231} = 0 \ ,$$
$$R^0_{232} = 0 \ ,$$
$$R^0_{233} = 0 \ ,$$
$$R^0_{300} = 0 \ ,$$
$$R^0_{301} = 0 \ ,$$
$$R^0_{302} = 0 \ ,$$
$$R^0_{303} = -\frac{0.5r\sin^2(\theta)\frac{d}{dr}A(r)}{A(r)B(r)} \ ,$$
$$R^0_{310} = 0 \ ,$$
$$R^0_{311} = 0 \ ,$$

$$R^0_{312} = 0 \ ,$$
$$R^0_{313} = 0 \ ,$$
$$R^0_{320} = 0 \ ,$$
$$R^0_{321} = 0 \ ,$$
$$R^0_{322} = 0 \ ,$$
$$R^0_{323} = 0 \ ,$$
$$R^0_{330} = \frac{0.5r\sin^2(\theta)\frac{d}{dr}A(r)}{A(r)B(r)} \ ,$$
$$R^0_{331} = 0 \ ,$$
$$R^0_{332} = 0 \ ,$$
$$R^0_{333} = 0 \ ,$$
$$R^1_{000} = 0 \ ,$$
$$R^1_{001} = -\frac{0.5\frac{d^2}{dr^2}A(r)}{B(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{B^2(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A(r)B(r)} \ ,$$
$$R^1_{002} = 0 \ ,$$
$$R^1_{003} = 0 \ ,$$
$$R^1_{010} = \frac{0.5\frac{d^2}{dr^2}A(r)}{B(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{B^2(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A(r)B(r)} \ ,$$
$$R^1_{011} = 0 \ ,$$
$$R^1_{012} = 0 \ ,$$
$$R^1_{013} = 0 \ ,$$
$$R^1_{020} = 0 \ ,$$
$$R^1_{021} = 0 \ ,$$
$$R^1_{022} = 0 \ ,$$
$$R^1_{023} = 0 \ ,$$
$$R^1_{030} = 0 \ ,$$
$$R^1_{031} = 0 \ ,$$
$$R^1_{032} = 0 \ ,$$
$$R^1_{033} = 0 \ ,$$
$$R^1_{100} = 0 \ ,$$

$$R^1_{101} = 0 \ ,$$
$$R^1_{102} = 0 \ ,$$
$$R^1_{103} = 0 \ ,$$
$$R^1_{110} = 0 \ ,$$
$$R^1_{111} = 0 \ ,$$
$$R^1_{112} = 0 \ ,$$
$$R^1_{113} = 0 \ ,$$
$$R^1_{120} = 0 \ ,$$
$$R^1_{121} = 0 \ ,$$
$$R^1_{122} = 0 \ ,$$
$$R^1_{123} = 0 \ ,$$
$$R^1_{130} = 0 \ ,$$
$$R^1_{131} = 0 \ ,$$
$$R^1_{132} = 0 \ ,$$
$$R^1_{133} = 0 \ ,$$
$$R^1_{200} = 0 \ ,$$
$$R^1_{201} = 0 \ ,$$
$$R^1_{202} = 0 \ ,$$
$$R^1_{203} = 0 \ ,$$
$$R^1_{210} = 0 \ ,$$
$$R^1_{211} = 0 \ ,$$
$$R^1_{212} = \frac{0.5r\frac{d}{dr}B(r)}{B^2(r)} \ ,$$
$$R^1_{213} = 0 \ ,$$
$$R^1_{220} = 0 \ ,$$
$$R^1_{221} = -\frac{0.5r\frac{d}{dr}B(r)}{B^2(r)} \ ,$$
$$R^1_{222} = 0 \ ,$$
$$R^1_{223} = 0 \ ,$$
$$R^1_{230} = 0 \ ,$$

$$R^1_{231} = 0 \ ,$$
$$R^1_{232} = 0 \ ,$$
$$R^1_{233} = 0 \ ,$$
$$R^1_{300} = 0 \ ,$$
$$R^1_{301} = 0 \ ,$$
$$R^1_{302} = 0 \ ,$$
$$R^1_{303} = 0 \ ,$$
$$R^1_{310} = 0 \ ,$$
$$R^1_{311} = 0 \ ,$$
$$R^1_{312} = 0 \ ,$$
$$R^1_{313} = \frac{0.5r\sin^2(\theta)\frac{d}{dr}B(r)}{B^2(r)} \ ,$$
$$R^1_{320} = 0 \ ,$$
$$R^1_{321} = 0 \ ,$$
$$R^1_{322} = 0 \ ,$$
$$R^1_{323} = 0 \ ,$$
$$R^1_{330} = 0 \ ,$$
$$R^1_{331} = -\frac{0.5r\sin^2(\theta)\frac{d}{dr}B(r)}{B^2(r)} \ ,$$
$$R^1_{332} = 0 \ ,$$
$$R^1_{333} = 0 \ ,$$
$$R^2_{000} = 0 \ ,$$
$$R^2_{001} = 0 \ ,$$
$$R^2_{002} = -\frac{0.5\frac{d}{dr}A(r)}{rB(r)} \ ,$$
$$R^2_{003} = 0 \ ,$$
$$R^2_{010} = 0 \ ,$$
$$R^2_{011} = 0 \ ,$$
$$R^2_{012} = 0 \ ,$$
$$R^2_{013} = 0 \ ,$$

$$R^2_{020} = \frac{0.5\frac{d}{dr}A(r)}{rB(r)} \ ,$$
$$R^2_{021} = 0 \ ,$$
$$R^2_{022} = 0 \ ,$$
$$R^2_{023} = 0 \ ,$$
$$R^2_{030} = 0 \ ,$$
$$R^2_{031} = 0 \ ,$$
$$R^2_{032} = 0 \ ,$$
$$R^2_{033} = 0 \ ,$$
$$R^2_{100} = 0 \ ,$$
$$R^2_{101} = 0 \ ,$$
$$R^2_{102} = 0 \ ,$$
$$R^2_{103} = 0 \ ,$$
$$R^2_{110} = 0 \ ,$$
$$R^2_{111} = 0 \ ,$$
$$R^2_{112} = -\frac{0.5\frac{d}{dr}B(r)}{rB(r)} \ ,$$
$$R^2_{113} = 0 \ ,$$
$$R^2_{120} = 0 \ ,$$
$$R^2_{121} = \frac{0.5\frac{d}{dr}B(r)}{rB(r)} \ ,$$
$$R^2_{122} = 0 \ ,$$
$$R^2_{123} = 0 \ ,$$
$$R^2_{130} = 0 \ ,$$
$$R^2_{131} = 0 \ ,$$
$$R^2_{132} = 0 \ ,$$
$$R^2_{133} = 0 \ ,$$
$$R^2_{200} = 0 \ ,$$
$$R^2_{201} = 0 \ ,$$
$$R^2_{202} = 0 \ ,$$
$$R^2_{203} = 0 \ ,$$

$$R^2_{210} = 0 \ ,$$
$$R^2_{211} = 0 \ ,$$
$$R^2_{212} = 0 \ ,$$
$$R^2_{213} = 0 \ ,$$
$$R^2_{220} = 0 \ ,$$
$$R^2_{221} = 0 \ ,$$
$$R^2_{222} = 0 \ ,$$
$$R^2_{223} = 0 \ ,$$
$$R^2_{230} = 0 \ ,$$
$$R^2_{231} = 0 \ ,$$
$$R^2_{232} = 0 \ ,$$
$$R^2_{233} = 0 \ ,$$
$$R^2_{300} = 0 \ ,$$
$$R^2_{301} = 0 \ ,$$
$$R^2_{302} = 0 \ ,$$
$$R^2_{303} = 0 \ ,$$
$$R^2_{310} = 0 \ ,$$
$$R^2_{311} = 0 \ ,$$
$$R^2_{312} = 0 \ ,$$
$$R^2_{313} = 0 \ ,$$
$$R^2_{320} = 0 \ ,$$
$$R^2_{321} = 0 \ ,$$
$$R^2_{322} = 0 \ ,$$
$$R^2_{323} = \frac{1.0\,(B(r) - 1)\sin^2(\theta)}{B(r)} \ ,$$
$$R^2_{330} = 0 \ ,$$
$$R^2_{331} = 0 \ ,$$
$$R^2_{332} = \frac{1.0 \cdot (1 - B(r))\sin^2(\theta)}{B(r)} \ ,$$
$$R^2_{333} = 0 \ ,$$
$$R^3_{000} = 0 \ ,$$

$$R^3_{001} = 0 \ ,$$
$$R^3_{002} = 0 \ ,$$
$$R^3_{003} = -\frac{0.5\frac{d}{dr}A(r)}{rB(r)} \ ,$$
$$R^3_{010} = 0 \ ,$$
$$R^3_{011} = 0 \ ,$$
$$R^3_{012} = 0 \ ,$$
$$R^3_{013} = 0 \ ,$$
$$R^3_{020} = 0 \ ,$$
$$R^3_{021} = 0 \ ,$$
$$R^3_{022} = 0 \ ,$$
$$R^3_{023} = 0 \ ,$$
$$R^3_{030} = \frac{0.5\frac{d}{dr}A(r)}{rB(r)} \ ,$$
$$R^3_{031} = 0 \ ,$$
$$R^3_{032} = 0 \ ,$$
$$R^3_{033} = 0 \ ,$$
$$R^3_{100} = 0 \ ,$$
$$R^3_{101} = 0 \ ,$$
$$R^3_{102} = 0 \ ,$$
$$R^3_{103} = 0 \ ,$$
$$R^3_{110} = 0 \ ,$$
$$R^3_{111} = 0 \ ,$$
$$R^3_{112} = 0 \ ,$$
$$R^3_{113} = -\frac{0.5\frac{d}{dr}B(r)}{rB(r)} \ ,$$
$$R^3_{120} = 0 \ ,$$
$$R^3_{121} = 0 \ ,$$
$$R^3_{122} = 0 \ ,$$
$$R^3_{123} = 0 \ ,$$
$$R^3_{130} = 0 \ ,$$

$$R^3_{131} = \frac{0.5\frac{d}{dr}B(r)}{rB(r)} \ ,$$
$$R^3_{132} = 0 \ ,$$
$$R^3_{133} = 0 \ ,$$
$$R^3_{200} = 0 \ ,$$
$$R^3_{201} = 0 \ ,$$
$$R^3_{202} = 0 \ ,$$
$$R^3_{203} = 0 \ ,$$
$$R^3_{210} = 0 \ ,$$
$$R^3_{211} = 0 \ ,$$
$$R^3_{212} = 0 \ ,$$
$$R^3_{213} = 0 \ ,$$
$$R^3_{220} = 0 \ ,$$
$$R^3_{221} = 0 \ ,$$
$$R^3_{222} = 0 \ ,$$
$$R^3_{223} = -1.0 + \frac{1.0}{B(r)} \ ,$$
$$R^3_{230} = 0 \ ,$$
$$R^3_{231} = 0 \ ,$$
$$R^3_{232} = 1.0 - \frac{1.0}{B(r)} \ ,$$
$$R^3_{233} = 0 \ ,$$
$$R^3_{300} = 0 \ ,$$
$$R^3_{301} = 0 \ ,$$
$$R^3_{302} = 0 \ ,$$
$$R^3_{303} = 0 \ ,$$
$$R^3_{310} = 0 \ ,$$
$$R^3_{311} = 0 \ ,$$
$$R^3_{312} = 0 \ ,$$
$$R^3_{313} = 0 \ ,$$
$$R^3_{320} = 0 \ ,$$

$$R^3_{321} = 0 \ ,$$
$$R^3_{322} = 0 \ ,$$
$$R^3_{323} = 0 \ ,$$
$$R^3_{330} = 0 \ ,$$
$$R^3_{331} = 0 \ ,$$
$$R^3_{332} = 0 \ ,$$
$$R^3_{333} = 0 \ ,$$
$$.$$

1.3 Ricci tensor

The Ricci tensor is

$$R_{00} = \frac{0.5\frac{d^2}{dr^2}A(r)}{B(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{B^2(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A(r)B(r)} + \frac{1.0\frac{d}{dr}A(r)}{rB(r)} \ ,$$
$$R_{01} = 0 \ ,$$
$$R_{02} = 0 \ ,$$
$$R_{03} = 0 \ ,$$
$$R_{10} = 0 \ ,$$
$$R_{11} = -\frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)} + \frac{1.0\frac{d}{dr}B(r)}{rB(r)} \ ,$$
$$R_{12} = 0 \ ,$$
$$R_{13} = 0 \ ,$$
$$R_{20} = 0 \ ,$$
$$R_{21} = 0 \ ,$$
$$R_{22} = \frac{0.5r\frac{d}{dr}B(r)}{B^2(r)} - \frac{0.5r\frac{d}{dr}A(r)}{A(r)B(r)} + 1.0 - \frac{1.0}{B(r)} \ ,$$
$$R_{23} = 0 \ ,$$
$$R_{30} = 0 \ ,$$
$$R_{31} = 0 \ ,$$
$$R_{32} = 0 \ ,$$
$$R_{33} = \frac{\left(0.5rA(r)\frac{d}{dr}B(r) - 0.5rB(r)\frac{d}{dr}A(r) + 1.0A(r)B^2(r) - 1.0A(r)B(r)\right)\sin^2(\theta)}{A(r)B^2(r)} \ ,$$
$$.$$

1.4 Ricci scalar

The Ricci scalar is

$$R = -\frac{1.0\frac{d^2}{dr^2}A(r)}{A(r)B(r)} + \frac{0.5\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B^2(r)} + \frac{0.5\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)B(r)} + \frac{2.0\frac{d}{dr}B(r)}{rB^2(r)} - \frac{2.0\frac{d}{dr}A(r)}{rA(r)B(r)} + \frac{2.0}{r^2} - \frac{2.0}{r^2B(r)} .$$

1.5 Einstein tensor

The Einstein tensor is

$$G_{00} = \frac{1.0\frac{d^2}{dr^2}A(r)}{B(r)} - \frac{0.5\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{B^2(r)} - \frac{0.5\left(\frac{d}{dr}A(r)\right)^2}{A(r)B(r)} - \frac{1.0A(r)\frac{d}{dr}B(r)}{rB^2(r)} + \frac{2.0\frac{d}{dr}A(r)}{rB(r)} - \frac{1.0A(r)}{r^2} + \frac{1.0A(r)}{r^2B(r)}$$

$$G_{01} = 0 ,$$

$$G_{02} = 0 ,$$

$$G_{03} = 0 ,$$

$$G_{10} = 0 ,$$

$$G_{11} = -\frac{1.0\frac{d^2}{dr^2}A(r)}{A(r)} + \frac{0.5\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B(r)} + \frac{0.5\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)} + \frac{2.0\frac{d}{dr}B(r)}{rB(r)} - \frac{1.0\frac{d}{dr}A(r)}{rA(r)} + \frac{1.0B(r)}{r^2} - \frac{1.0}{r^2} ,$$

$$G_{12} = 0 ,$$

$$G_{13} = 0 ,$$

$$G_{20} = 0 ,$$

$$G_{21} = 0 ,$$

$$G_{22} = -\frac{0.5r^2\frac{d^2}{dr^2}A(r)}{A(r)B(r)} + \frac{0.25r^2\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B^2(r)} + \frac{0.25r^2\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)B(r)} + \frac{1.5r\frac{d}{dr}B(r)}{B^2(r)} - \frac{1.5r\frac{d}{dr}A(r)}{A(r)B(r)} + 2.0 - \frac{2.0}{B(r)}$$

$$G_{23} = 0 ,$$

$$G_{30} = 0 ,$$

$$G_{31} = 0 ,$$

$$G_{32} = 0 ,$$

$$G_{33} = \frac{\left(-0.5r^2A(r)B(r)\frac{d^2}{dr^2}A(r) + 0.25r^2A(r)\frac{d}{dr}A(r)\frac{d}{dr}B(r) + 0.25r^2B(r)\left(\frac{d}{dr}A(r)\right)^2 + 1.5rA^2(r)\frac{d}{dr}B(\right.}{A^2(r)B^2(r)}$$

.

Chapter 2

Interior Schwarschild metric

The metric is

$$g = \begin{bmatrix} -e^{\nu(r)} & 0 & 0 & 0 \\ 0 & e^{\lambda(r)} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix} ,$$

.

2.1 Christoffel symbols

The Christoffel symbols are

$$\Gamma^0_{00} = 0 ,$$

$$\Gamma^0_{01} = 0.5 \frac{d}{dr} \nu(r) ,$$

$$\Gamma^0_{02} = 0 ,$$

$$\Gamma^0_{03} = 0 ,$$

$$\Gamma^0_{10} = 0.5 \frac{d}{dr} \nu(r) ,$$

$$\Gamma^0_{11} = 0 ,$$

$$\Gamma^0_{12} = 0 ,$$

$$\Gamma^0_{13} = 0 ,$$

$$\Gamma^0_{20} = 0 ,$$

$$\Gamma^0_{21} = 0 ,$$

$$\Gamma^0_{22} = 0 ,$$

$$\Gamma^0_{23} = 0 ,$$

$$\Gamma^0_{30} = 0 ,$$

$$\Gamma^0_{31} = 0 \ ,$$
$$\Gamma^0_{32} = 0 \ ,$$
$$\Gamma^0_{33} = 0 \ ,$$
$$\Gamma^1_{00} = 0.5 e^{-\lambda(r)+\nu(r)} \frac{d}{dr}\nu(r) \ ,$$
$$\Gamma^1_{01} = 0 \ ,$$
$$\Gamma^1_{02} = 0 \ ,$$
$$\Gamma^1_{03} = 0 \ ,$$
$$\Gamma^1_{10} = 0 \ ,$$
$$\Gamma^1_{11} = 0.5 \frac{d}{dr}\lambda(r) \ ,$$
$$\Gamma^1_{12} = 0 \ ,$$
$$\Gamma^1_{13} = 0 \ ,$$
$$\Gamma^1_{20} = 0 \ ,$$
$$\Gamma^1_{21} = 0 \ ,$$
$$\Gamma^1_{22} = -1.0 r e^{-\lambda(r)} \ ,$$
$$\Gamma^1_{23} = 0 \ ,$$
$$\Gamma^1_{30} = 0 \ ,$$
$$\Gamma^1_{31} = 0 \ ,$$
$$\Gamma^1_{32} = 0 \ ,$$
$$\Gamma^1_{33} = -1.0 r e^{-\lambda(r)} \sin^2(\theta) \ ,$$
$$\Gamma^2_{00} = 0 \ ,$$
$$\Gamma^2_{01} = 0 \ ,$$
$$\Gamma^2_{02} = 0 \ ,$$
$$\Gamma^2_{03} = 0 \ ,$$
$$\Gamma^2_{10} = 0 \ ,$$
$$\Gamma^2_{11} = 0 \ ,$$
$$\Gamma^2_{12} = \frac{1.0}{r} \ ,$$
$$\Gamma^2_{13} = 0 \ ,$$
$$\Gamma^2_{20} = 0 \ ,$$

$$\Gamma^2_{21} = \frac{1.0}{r} \ ,$$

$$\Gamma^2_{22} = 0 \ ,$$

$$\Gamma^2_{23} = 0 \ ,$$

$$\Gamma^2_{30} = 0 \ ,$$

$$\Gamma^2_{31} = 0 \ ,$$

$$\Gamma^2_{32} = 0 \ ,$$

$$\Gamma^2_{33} = -0.5 \sin(2\theta) \ ,$$

$$\Gamma^3_{00} = 0 \ ,$$

$$\Gamma^3_{01} = 0 \ ,$$

$$\Gamma^3_{02} = 0 \ ,$$

$$\Gamma^3_{03} = 0 \ ,$$

$$\Gamma^3_{10} = 0 \ ,$$

$$\Gamma^3_{11} = 0 \ ,$$

$$\Gamma^3_{12} = 0 \ ,$$

$$\Gamma^3_{13} = \frac{1.0}{r} \ ,$$

$$\Gamma^3_{20} = 0 \ ,$$

$$\Gamma^3_{21} = 0 \ ,$$

$$\Gamma^3_{22} = 0 \ ,$$

$$\Gamma^3_{23} = \frac{1.0}{\tan(\theta)} \ ,$$

$$\Gamma^3_{30} = 0 \ ,$$

$$\Gamma^3_{31} = \frac{1.0}{r} \ ,$$

$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)} \ ,$$

$$\Gamma^3_{33} = 0 \ ,$$

.

2.2 Riemann tensor

The Riemann tensor is

$$R^0_{000} = 0 \ ,$$
$$R^0_{001} = 0 \ ,$$
$$R^0_{002} = 0 \ ,$$
$$R^0_{003} = 0 \ ,$$
$$R^0_{010} = 0 \ ,$$
$$R^0_{011} = 0 \ ,$$
$$R^0_{012} = 0 \ ,$$
$$R^0_{013} = 0 \ ,$$
$$R^0_{020} = 0 \ ,$$
$$R^0_{021} = 0 \ ,$$
$$R^0_{022} = 0 \ ,$$
$$R^0_{023} = 0 \ ,$$
$$R^0_{030} = 0 \ ,$$
$$R^0_{031} = 0 \ ,$$
$$R^0_{032} = 0 \ ,$$
$$R^0_{033} = 0 \ ,$$
$$R^0_{100} = 0 \ ,$$
$$R^0_{101} = 0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.25\left(\frac{d}{dr}\nu(r)\right)^2 - 0.5\frac{d^2}{dr^2}\nu(r) \ ,$$
$$R^0_{102} = 0 \ ,$$
$$R^0_{103} = 0 \ ,$$
$$R^0_{110} = -0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5\frac{d^2}{dr^2}\nu(r) \ ,$$
$$R^0_{111} = 0 \ ,$$
$$R^0_{112} = 0 \ ,$$
$$R^0_{113} = 0 \ ,$$
$$R^0_{120} = 0 \ ,$$
$$R^0_{121} = 0 \ ,$$

$$R^0_{122} = 0 \ ,$$
$$R^0_{123} = 0 \ ,$$
$$R^0_{130} = 0 \ ,$$
$$R^0_{131} = 0 \ ,$$
$$R^0_{132} = 0 \ ,$$
$$R^0_{133} = 0 \ ,$$
$$R^0_{200} = 0 \ ,$$
$$R^0_{201} = 0 \ ,$$
$$R^0_{202} = -0.5re^{-\lambda(r)}\frac{d}{dr}\nu(r) \ ,$$
$$R^0_{203} = 0 \ ,$$
$$R^0_{210} = 0 \ ,$$
$$R^0_{211} = 0 \ ,$$
$$R^0_{212} = 0 \ ,$$
$$R^0_{213} = 0 \ ,$$
$$R^0_{220} = 0.5re^{-\lambda(r)}\frac{d}{dr}\nu(r) \ ,$$
$$R^0_{221} = 0 \ ,$$
$$R^0_{222} = 0 \ ,$$
$$R^0_{223} = 0 \ ,$$
$$R^0_{230} = 0 \ ,$$
$$R^0_{231} = 0 \ ,$$
$$R^0_{232} = 0 \ ,$$
$$R^0_{233} = 0 \ ,$$
$$R^0_{300} = 0 \ ,$$
$$R^0_{301} = 0 \ ,$$
$$R^0_{302} = 0 \ ,$$
$$R^0_{303} = -0.5re^{-\lambda(r)}\sin^2(\theta)\frac{d}{dr}\nu(r) \ ,$$
$$R^0_{310} = 0 \ ,$$
$$R^0_{311} = 0 \ ,$$

$$R^0_{312} = 0 \ ,$$
$$R^0_{313} = 0 \ ,$$
$$R^0_{320} = 0 \ ,$$
$$R^0_{321} = 0 \ ,$$
$$R^0_{322} = 0 \ ,$$
$$R^0_{323} = 0 \ ,$$
$$R^0_{330} = 0.5re^{-\lambda(r)} \sin^2(\theta) \frac{d}{dr}\nu(r) \ ,$$
$$R^0_{331} = 0 \ ,$$
$$R^0_{332} = 0 \ ,$$
$$R^0_{333} = 0 \ ,$$
$$R^1_{000} = 0 \ ,$$
$$R^1_{001} = \left(0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.25\left(\frac{d}{dr}\nu(r)\right)^2 - 0.5\frac{d^2}{dr^2}\nu(r)\right) e^{-\lambda(r)+\nu(r)} \ ,$$
$$R^1_{002} = 0 \ ,$$
$$R^1_{003} = 0 \ ,$$
$$R^1_{010} = \left(-0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5\frac{d^2}{dr^2}\nu(r)\right) e^{-\lambda(r)+\nu(r)} \ ,$$
$$R^1_{011} = 0 \ ,$$
$$R^1_{012} = 0 \ ,$$
$$R^1_{013} = 0 \ ,$$
$$R^1_{020} = 0 \ ,$$
$$R^1_{021} = 0 \ ,$$
$$R^1_{022} = 0 \ ,$$
$$R^1_{023} = 0 \ ,$$
$$R^1_{030} = 0 \ ,$$
$$R^1_{031} = 0 \ ,$$
$$R^1_{032} = 0 \ ,$$
$$R^1_{033} = 0 \ ,$$
$$R^1_{100} = 0 \ ,$$

$$R^1_{101} = 0 \ ,$$
$$R^1_{102} = 0 \ ,$$
$$R^1_{103} = 0 \ ,$$
$$R^1_{110} = 0 \ ,$$
$$R^1_{111} = 0 \ ,$$
$$R^1_{112} = 0 \ ,$$
$$R^1_{113} = 0 \ ,$$
$$R^1_{120} = 0 \ ,$$
$$R^1_{121} = 0 \ ,$$
$$R^1_{122} = 0 \ ,$$
$$R^1_{123} = 0 \ ,$$
$$R^1_{130} = 0 \ ,$$
$$R^1_{131} = 0 \ ,$$
$$R^1_{132} = 0 \ ,$$
$$R^1_{133} = 0 \ ,$$
$$R^1_{200} = 0 \ ,$$
$$R^1_{201} = 0 \ ,$$
$$R^1_{202} = 0 \ ,$$
$$R^1_{203} = 0 \ ,$$
$$R^1_{210} = 0 \ ,$$
$$R^1_{211} = 0 \ ,$$
$$R^1_{212} = 0.5re^{-\lambda(r)}\frac{d}{dr}\lambda(r) \ ,$$
$$R^1_{213} = 0 \ ,$$
$$R^1_{220} = 0 \ ,$$
$$R^1_{221} = -0.5re^{-\lambda(r)}\frac{d}{dr}\lambda(r) \ ,$$
$$R^1_{222} = 0 \ ,$$
$$R^1_{223} = 0 \ ,$$
$$R^1_{230} = 0 \ ,$$
$$R^1_{231} = 0 \ ,$$

$$R^1_{232} = 0 \ ,$$
$$R^1_{233} = 0 \ ,$$
$$R^1_{300} = 0 \ ,$$
$$R^1_{301} = 0 \ ,$$
$$R^1_{302} = 0 \ ,$$
$$R^1_{303} = 0 \ ,$$
$$R^1_{310} = 0 \ ,$$
$$R^1_{311} = 0 \ ,$$
$$R^1_{312} = 0 \ ,$$
$$R^1_{313} = 0.5re^{-\lambda(r)} \sin^2(\theta) \frac{d}{dr}\lambda(r) \ ,$$
$$R^1_{320} = 0 \ ,$$
$$R^1_{321} = 0 \ ,$$
$$R^1_{322} = 0 \ ,$$
$$R^1_{323} = 0 \ ,$$
$$R^1_{330} = 0 \ ,$$
$$R^1_{331} = -0.5re^{-\lambda(r)} \sin^2(\theta) \frac{d}{dr}\lambda(r) \ ,$$
$$R^1_{332} = 0 \ ,$$
$$R^1_{333} = 0 \ ,$$
$$R^2_{000} = 0 \ ,$$
$$R^2_{001} = 0 \ ,$$
$$R^2_{002} = -\frac{0.5e^{-\lambda(r)+\nu(r)} \frac{d}{dr}\nu(r)}{r} \ ,$$
$$R^2_{003} = 0 \ ,$$
$$R^2_{010} = 0 \ ,$$
$$R^2_{011} = 0 \ ,$$
$$R^2_{012} = 0 \ ,$$
$$R^2_{013} = 0 \ ,$$
$$R^2_{020} = \frac{0.5e^{-\lambda(r)+\nu(r)} \frac{d}{dr}\nu(r)}{r} \ ,$$
$$R^2_{021} = 0 \ ,$$

$$R^2_{022} = 0 \ ,$$
$$R^2_{023} = 0 \ ,$$
$$R^2_{030} = 0 \ ,$$
$$R^2_{031} = 0 \ ,$$
$$R^2_{032} = 0 \ ,$$
$$R^2_{033} = 0 \ ,$$
$$R^2_{100} = 0 \ ,$$
$$R^2_{101} = 0 \ ,$$
$$R^2_{102} = 0 \ ,$$
$$R^2_{103} = 0 \ ,$$
$$R^2_{110} = 0 \ ,$$
$$R^2_{111} = 0 \ ,$$
$$R^2_{112} = -\frac{0.5\frac{d}{dr}\lambda(r)}{r} \ ,$$
$$R^2_{113} = 0 \ ,$$
$$R^2_{120} = 0 \ ,$$
$$R^2_{121} = \frac{0.5\frac{d}{dr}\lambda(r)}{r} \ ,$$
$$R^2_{122} = 0 \ ,$$
$$R^2_{123} = 0 \ ,$$
$$R^2_{130} = 0 \ ,$$
$$R^2_{131} = 0 \ ,$$
$$R^2_{132} = 0 \ ,$$
$$R^2_{133} = 0 \ ,$$
$$R^2_{200} = 0 \ ,$$
$$R^2_{201} = 0 \ ,$$
$$R^2_{202} = 0 \ ,$$
$$R^2_{203} = 0 \ ,$$
$$R^2_{210} = 0 \ ,$$
$$R^2_{211} = 0 \ ,$$
$$R^2_{212} = 0 \ ,$$

$$R^2_{213} = 0 \ ,$$
$$R^2_{220} = 0 \ ,$$
$$R^2_{221} = 0 \ ,$$
$$R^2_{222} = 0 \ ,$$
$$R^2_{223} = 0 \ ,$$
$$R^2_{230} = 0 \ ,$$
$$R^2_{231} = 0 \ ,$$
$$R^2_{232} = 0 \ ,$$
$$R^2_{233} = 0 \ ,$$
$$R^2_{300} = 0 \ ,$$
$$R^2_{301} = 0 \ ,$$
$$R^2_{302} = 0 \ ,$$
$$R^2_{303} = 0 \ ,$$
$$R^2_{310} = 0 \ ,$$
$$R^2_{311} = 0 \ ,$$
$$R^2_{312} = 0 \ ,$$
$$R^2_{313} = 0 \ ,$$
$$R^2_{320} = 0 \ ,$$
$$R^2_{321} = 0 \ ,$$
$$R^2_{322} = 0 \ ,$$
$$R^2_{323} = 1.0 \left(e^{\lambda(r)} - 1\right) e^{-\lambda(r)} \sin^2(\theta) \ ,$$
$$R^2_{330} = 0 \ ,$$
$$R^2_{331} = 0 \ ,$$
$$R^2_{332} = 1.0 \cdot \left(1 - e^{\lambda(r)}\right) e^{-\lambda(r)} \sin^2(\theta) \ ,$$
$$R^2_{333} = 0 \ ,$$
$$R^3_{000} = 0 \ ,$$
$$R^3_{001} = 0 \ ,$$
$$R^3_{002} = 0 \ ,$$
$$R^3_{003} = -\frac{0.5 e^{-\lambda(r)+\nu(r)} \frac{d}{dr} \nu(r)}{r} \ ,$$

$$R^3_{010} = 0 \ ,$$
$$R^3_{011} = 0 \ ,$$
$$R^3_{012} = 0 \ ,$$
$$R^3_{013} = 0 \ ,$$
$$R^3_{020} = 0 \ ,$$
$$R^3_{021} = 0 \ ,$$
$$R^3_{022} = 0 \ ,$$
$$R^3_{023} = 0 \ ,$$
$$R^3_{030} = \frac{0.5e^{-\lambda(r)+\nu(r)}\frac{d}{dr}\nu(r)}{r} \ ,$$
$$R^3_{031} = 0 \ ,$$
$$R^3_{032} = 0 \ ,$$
$$R^3_{033} = 0 \ ,$$
$$R^3_{100} = 0 \ ,$$
$$R^3_{101} = 0 \ ,$$
$$R^3_{102} = 0 \ ,$$
$$R^3_{103} = 0 \ ,$$
$$R^3_{110} = 0 \ ,$$
$$R^3_{111} = 0 \ ,$$
$$R^3_{112} = 0 \ ,$$
$$R^3_{113} = -\frac{0.5\frac{d}{dr}\lambda(r)}{r} \ ,$$
$$R^3_{120} = 0 \ ,$$
$$R^3_{121} = 0 \ ,$$
$$R^3_{122} = 0 \ ,$$
$$R^3_{123} = 0 \ ,$$
$$R^3_{130} = 0 \ ,$$
$$R^3_{131} = \frac{0.5\frac{d}{dr}\lambda(r)}{r} \ ,$$
$$R^3_{132} = 0 \ ,$$
$$R^3_{133} = 0 \ ,$$

$$
\begin{aligned}
R^3_{200} &= 0 \ , \\
R^3_{201} &= 0 \ , \\
R^3_{202} &= 0 \ , \\
R^3_{203} &= 0 \ , \\
R^3_{210} &= 0 \ , \\
R^3_{211} &= 0 \ , \\
R^3_{212} &= 0 \ , \\
R^3_{213} &= 0 \ , \\
R^3_{220} &= 0 \ , \\
R^3_{221} &= 0 \ , \\
R^3_{222} &= 0 \ , \\
R^3_{223} &= -1.0 + 1.0e^{-\lambda(r)} \ , \\
R^3_{230} &= 0 \ , \\
R^3_{231} &= 0 \ , \\
R^3_{232} &= 1.0 - 1.0e^{-\lambda(r)} \ , \\
R^3_{233} &= 0 \ , \\
R^3_{300} &= 0 \ , \\
R^3_{301} &= 0 \ , \\
R^3_{302} &= 0 \ , \\
R^3_{303} &= 0 \ , \\
R^3_{310} &= 0 \ , \\
R^3_{311} &= 0 \ , \\
R^3_{312} &= 0 \ , \\
R^3_{313} &= 0 \ , \\
R^3_{320} &= 0 \ , \\
R^3_{321} &= 0 \ , \\
R^3_{322} &= 0 \ , \\
R^3_{323} &= 0 \ , \\
R^3_{330} &= 0 \ , \\
R^3_{331} &= 0 \ , \\
R^3_{332} &= 0 \ , \\
R^3_{333} &= 0 \ ,
\end{aligned}
$$

.

2.3 Ricci tensor

The Ricci tensor is

$$R_{00} = \frac{\left(-0.25r\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) + 0.25r\left(\frac{d}{dr}\nu(r)\right)^2 + 0.5r\frac{d^2}{dr^2}\nu(r) + 1.0\frac{d}{dr}\nu(r)\right)e^{-\lambda(r)+\nu(r)}}{r} ,$$

$$R_{01} = 0 ,$$

$$R_{02} = 0 ,$$

$$R_{03} = 0 ,$$

$$R_{10} = 0 ,$$

$$R_{11} = 0.25\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.25\left(\frac{d}{dr}\nu(r)\right)^2 - 0.5\frac{d^2}{dr^2}\nu(r) + \frac{1.0\frac{d}{dr}\lambda(r)}{r} ,$$

$$R_{12} = 0 ,$$

$$R_{13} = 0 ,$$

$$R_{20} = 0 ,$$

$$R_{21} = 0 ,$$

$$R_{22} = \left(0.5r\left(\frac{d}{dr}\lambda(r) - \frac{d}{dr}\nu(r)\right) + 1.0e^{\lambda(r)} - 1.0\right)e^{-\lambda(r)} ,$$

$$R_{23} = 0 ,$$

$$R_{30} = 0 ,$$

$$R_{31} = 0 ,$$

$$R_{32} = 0 ,$$

$$R_{33} = \left(0.5r\frac{d}{dr}\lambda(r) - 0.5r\frac{d}{dr}\nu(r) + 1.0e^{\lambda(r)} - 1.0\right)e^{-\lambda(r)}\sin^2(\theta) ,$$

.

2.4 Ricci scalar

The Ricci scalar is

$$R = \frac{\left(0.5r^2\frac{d}{dr}\lambda(r)\frac{d}{dr}\nu(r) - 0.5r^2\left(\frac{d}{dr}\nu(r)\right)^2 - 1.0r^2\frac{d^2}{dr^2}\nu(r) + 2.0r\frac{d}{dr}\lambda(r) - 2.0r\frac{d}{dr}\nu(r) + 2.0e^{\lambda(r)} - 2.0\right)}{r^2}$$

2.5 Einstein tensor

The Einstein tensor is

$$G_{00} = \frac{1.0\frac{d^2}{dr^2}A(r)}{B(r)} - \frac{0.5\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{B^2(r)} - \frac{0.5\left(\frac{d}{dr}A(r)\right)^2}{A(r)B(r)} - \frac{1.0A(r)\frac{d}{dr}B(r)}{rB^2(r)} + \frac{2.0\frac{d}{dr}A(r)}{rB(r)} - \frac{1.0A(r)}{r^2} + \frac{1.0A(r)}{r^2B(r)}$$

$$G_{01} = 0 \ ,$$

$$G_{02} = 0 \ ,$$

$$G_{03} = 0 \ ,$$

$$G_{10} = 0 \ ,$$

$$G_{11} = -\frac{1.0\frac{d^2}{dr^2}A(r)}{A(r)} + \frac{0.5\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B(r)} + \frac{0.5\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)} + \frac{2.0\frac{d}{dr}B(r)}{rB(r)} - \frac{1.0\frac{d}{dr}A(r)}{rA(r)} + \frac{1.0B(r)}{r^2} - \frac{1.0}{r^2} \ ,$$

$$G_{12} = 0 \ ,$$

$$G_{13} = 0 \ ,$$

$$G_{20} = 0 \ ,$$

$$G_{21} = 0 \ ,$$

$$G_{22} = -\frac{0.5r^2\frac{d^2}{dr^2}A(r)}{A(r)B(r)} + \frac{0.25r^2\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B^2(r)} + \frac{0.25r^2\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)B(r)} + \frac{1.5r\frac{d}{dr}B(r)}{B^2(r)} - \frac{1.5r\frac{d}{dr}A(r)}{A(r)B(r)} + 2.0 - \frac{2.0}{B(r)}$$

$$G_{23} = 0 \ ,$$

$$G_{30} = 0 \ ,$$

$$G_{31} = 0 \ ,$$

$$G_{32} = 0 \ ,$$

$$G_{33} = \frac{\left(-0.5r^2A(r)B(r)\frac{d^2}{dr^2}A(r) + 0.25r^2A(r)\frac{d}{dr}A(r)\frac{d}{dr}B(r) + 0.25r^2B(r)\left(\frac{d}{dr}A(r)\right)^2 + 1.5rA^2(r)\frac{d}{dr}B(\right.}{A^2(r)B^2(r)}$$

.

Chapter 3

FRW metric

The metric is

$$g = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -\frac{a^2(t)}{kr^2-1} & 0 & 0 \\ 0 & 0 & r^2a^2(t) & 0 \\ 0 & 0 & 0 & r^2a^2(t)\sin^2(\theta) \end{bmatrix} ,$$

.

3.1 Christoffel symbols

The Christoffel symbols are

$$\Gamma^0_{00} = 0 ,$$

$$\Gamma^0_{01} = 0 ,$$

$$\Gamma^0_{02} = 0 ,$$

$$\Gamma^0_{03} = 0 ,$$

$$\Gamma^0_{10} = 0 ,$$

$$\Gamma^0_{11} = -\frac{1.0a(t)\frac{d}{dt}a(t)}{kr^2-1} ,$$

$$\Gamma^0_{12} = 0 ,$$

$$\Gamma^0_{13} = 0 ,$$

$$\Gamma^0_{20} = 0 ,$$

$$\Gamma^0_{21} = 0 ,$$

$$\Gamma^0_{22} = 1.0r^2a(t)\frac{d}{dt}a(t) ,$$

$$\Gamma^0_{23} = 0 \ ,$$
$$\Gamma^0_{30} = 0 \ ,$$
$$\Gamma^0_{31} = 0 \ ,$$
$$\Gamma^0_{32} = 0 \ ,$$
$$\Gamma^0_{33} = 1.0r^2 a(t) \sin^2(\theta) \frac{d}{dt} a(t) \ ,$$
$$\Gamma^1_{00} = 0 \ ,$$
$$\Gamma^1_{01} = \frac{1.0 \frac{d}{dt} a(t)}{a(t)} \ ,$$
$$\Gamma^1_{02} = 0 \ ,$$
$$\Gamma^1_{03} = 0 \ ,$$
$$\Gamma^1_{10} = \frac{1.0 \frac{d}{dt} a(t)}{a(t)} \ ,$$
$$\Gamma^1_{11} = -\frac{1.0kr}{kr^2 - 1} \ ,$$
$$\Gamma^1_{12} = 0 \ ,$$
$$\Gamma^1_{13} = 0 \ ,$$
$$\Gamma^1_{20} = 0 \ ,$$
$$\Gamma^1_{21} = 0 \ ,$$
$$\Gamma^1_{22} = 1.0r \left(kr^2 - 1\right) \ ,$$
$$\Gamma^1_{23} = 0 \ ,$$
$$\Gamma^1_{30} = 0 \ ,$$
$$\Gamma^1_{31} = 0 \ ,$$
$$\Gamma^1_{32} = 0 \ ,$$
$$\Gamma^1_{33} = 1.0r \left(kr^2 - 1\right) \sin^2(\theta) \ ,$$
$$\Gamma^2_{00} = 0 \ ,$$
$$\Gamma^2_{01} = 0 \ ,$$
$$\Gamma^2_{02} = \frac{1.0 \frac{d}{dt} a(t)}{a(t)} \ ,$$
$$\Gamma^2_{03} = 0 \ ,$$
$$\Gamma^2_{10} = 0 \ ,$$
$$\Gamma^2_{11} = 0 \ ,$$

$$\Gamma^2_{12} = \frac{1.0}{r} \ ,$$
$$\Gamma^2_{13} = 0 \ ,$$
$$\Gamma^2_{20} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} \ ,$$
$$\Gamma^2_{21} = \frac{1.0}{r} \ ,$$
$$\Gamma^2_{22} = 0 \ ,$$
$$\Gamma^2_{23} = 0 \ ,$$
$$\Gamma^2_{30} = 0 \ ,$$
$$\Gamma^2_{31} = 0 \ ,$$
$$\Gamma^2_{32} = 0 \ ,$$
$$\Gamma^2_{33} = -0.5\sin(2\theta) \ ,$$
$$\Gamma^3_{00} = 0 \ ,$$
$$\Gamma^3_{01} = 0 \ ,$$
$$\Gamma^3_{02} = 0 \ ,$$
$$\Gamma^3_{03} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} \ ,$$
$$\Gamma^3_{10} = 0 \ ,$$
$$\Gamma^3_{11} = 0 \ ,$$
$$\Gamma^3_{12} = 0 \ ,$$
$$\Gamma^3_{13} = \frac{1.0}{r} \ ,$$
$$\Gamma^3_{20} = 0 \ ,$$
$$\Gamma^3_{21} = 0 \ ,$$
$$\Gamma^3_{22} = 0 \ ,$$
$$\Gamma^3_{23} = \frac{1.0}{\tan(\theta)} \ ,$$
$$\Gamma^3_{30} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} \ ,$$
$$\Gamma^3_{31} = \frac{1.0}{r} \ ,$$
$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)} \ ,$$
$$\Gamma^3_{33} = 0 \ ,$$
$$.$$

3.2 Riemann tensor

The Riemann tensor is

$$R^0_{000} = 0 \ ,$$
$$R^0_{001} = 0 \ ,$$
$$R^0_{002} = 0 \ ,$$
$$R^0_{003} = 0 \ ,$$
$$R^0_{010} = 0 \ ,$$
$$R^0_{011} = 0 \ ,$$
$$R^0_{012} = 0 \ ,$$
$$R^0_{013} = 0 \ ,$$
$$R^0_{020} = 0 \ ,$$
$$R^0_{021} = 0 \ ,$$
$$R^0_{022} = 0 \ ,$$
$$R^0_{023} = 0 \ ,$$
$$R^0_{030} = 0 \ ,$$
$$R^0_{031} = 0 \ ,$$
$$R^0_{032} = 0 \ ,$$
$$R^0_{033} = 0 \ ,$$
$$R^0_{100} = 0 \ ,$$
$$R^0_{101} = -\frac{1.0a(t)\frac{d^2}{dt^2}a(t)}{kr^2 - 1} \ ,$$
$$R^0_{102} = 0 \ ,$$
$$R^0_{103} = 0 \ ,$$
$$R^0_{110} = \frac{1.0a(t)\frac{d^2}{dt^2}a(t)}{kr^2 - 1} \ ,$$
$$R^0_{111} = 0 \ ,$$
$$R^0_{112} = 0 \ ,$$
$$R^0_{113} = 0 \ ,$$
$$R^0_{120} = 0 \ ,$$
$$R^0_{121} = 0 \ ,$$

$$R^0_{122} = 0 \ ,$$
$$R^0_{123} = 0 \ ,$$
$$R^0_{130} = 0 \ ,$$
$$R^0_{131} = 0 \ ,$$
$$R^0_{132} = 0 \ ,$$
$$R^0_{133} = 0 \ ,$$
$$R^0_{200} = 0 \ ,$$
$$R^0_{201} = 0 \ ,$$
$$R^0_{202} = 1.0 r^2 a(t) \frac{d^2}{dt^2} a(t) \ ,$$
$$R^0_{203} = 0 \ ,$$
$$R^0_{210} = 0 \ ,$$
$$R^0_{211} = 0 \ ,$$
$$R^0_{212} = 0 \ ,$$
$$R^0_{213} = 0 \ ,$$
$$R^0_{220} = -1.0 r^2 a(t) \frac{d^2}{dt^2} a(t) \ ,$$
$$R^0_{221} = 0 \ ,$$
$$R^0_{222} = 0 \ ,$$
$$R^0_{223} = 0 \ ,$$
$$R^0_{230} = 0 \ ,$$
$$R^0_{231} = 0 \ ,$$
$$R^0_{232} = 0 \ ,$$
$$R^0_{233} = 0 \ ,$$
$$R^0_{300} = 0 \ ,$$
$$R^0_{301} = 0 \ ,$$
$$R^0_{302} = 0 \ ,$$
$$R^0_{303} = 1.0 r^2 a(t) \sin^2(\theta) \frac{d^2}{dt^2} a(t) \ ,$$
$$R^0_{310} = 0 \ ,$$
$$R^0_{311} = 0 \ ,$$

$$R^0_{312} = 0 \ ,$$
$$R^0_{313} = 0 \ ,$$
$$R^0_{320} = 0 \ ,$$
$$R^0_{321} = 0 \ ,$$
$$R^0_{322} = 0 \ ,$$
$$R^0_{323} = 0 \ ,$$
$$R^0_{330} = -1.0 r^2 a(t) \sin^2(\theta) \frac{d^2}{dt^2} a(t) \ ,$$
$$R^0_{331} = 0 \ ,$$
$$R^0_{332} = 0 \ ,$$
$$R^0_{333} = 0 \ ,$$
$$R^1_{000} = 0 \ ,$$
$$R^1_{001} = \frac{1.0 \frac{d^2}{dt^2} a(t)}{a(t)} \ ,$$
$$R^1_{002} = 0 \ ,$$
$$R^1_{003} = 0 \ ,$$
$$R^1_{010} = -\frac{1.0 \frac{d^2}{dt^2} a(t)}{a(t)} \ ,$$
$$R^1_{011} = 0 \ ,$$
$$R^1_{012} = 0 \ ,$$
$$R^1_{013} = 0 \ ,$$
$$R^1_{020} = 0 \ ,$$
$$R^1_{021} = 0 \ ,$$
$$R^1_{022} = 0 \ ,$$
$$R^1_{023} = 0 \ ,$$
$$R^1_{030} = 0 \ ,$$
$$R^1_{031} = 0 \ ,$$
$$R^1_{032} = 0 \ ,$$
$$R^1_{033} = 0 \ ,$$
$$R^1_{100} = 0 \ ,$$
$$R^1_{101} = 0 \ ,$$

$$R^1_{102} = 0 \ ,$$
$$R^1_{103} = 0 \ ,$$
$$R^1_{110} = 0 \ ,$$
$$R^1_{111} = 0 \ ,$$
$$R^1_{112} = 0 \ ,$$
$$R^1_{113} = 0 \ ,$$
$$R^1_{120} = 0 \ ,$$
$$R^1_{121} = 0 \ ,$$
$$R^1_{122} = 0 \ ,$$
$$R^1_{123} = 0 \ ,$$
$$R^1_{130} = 0 \ ,$$
$$R^1_{131} = 0 \ ,$$
$$R^1_{132} = 0 \ ,$$
$$R^1_{133} = 0 \ ,$$
$$R^1_{200} = 0 \ ,$$
$$R^1_{201} = 0 \ ,$$
$$R^1_{202} = 0 \ ,$$
$$R^1_{203} = 0 \ ,$$
$$R^1_{210} = 0 \ ,$$
$$R^1_{211} = 0 \ ,$$
$$R^1_{212} = 1.0r^2 \left(k + \left(\frac{d}{dt} a(t) \right)^2 \right) \ ,$$
$$R^1_{213} = 0 \ ,$$
$$R^1_{220} = 0 \ ,$$
$$R^1_{221} = 1.0r^2 \left(-k - \left(\frac{d}{dt} a(t) \right)^2 \right) \ ,$$
$$R^1_{222} = 0 \ ,$$
$$R^1_{223} = 0 \ ,$$
$$R^1_{230} = 0 \ ,$$
$$R^1_{231} = 0 \ ,$$

$$R^1_{232} = 0 \ ,$$
$$R^1_{233} = 0 \ ,$$
$$R^1_{300} = 0 \ ,$$
$$R^1_{301} = 0 \ ,$$
$$R^1_{302} = 0 \ ,$$
$$R^1_{303} = 0 \ ,$$
$$R^1_{310} = 0 \ ,$$
$$R^1_{311} = 0 \ ,$$
$$R^1_{312} = 0 \ ,$$
$$R^1_{313} = 1.0r^2 \left(k + \left(\frac{d}{dt}a(t)\right)^2\right) \sin^2(\theta) \ ,$$
$$R^1_{320} = 0 \ ,$$
$$R^1_{321} = 0 \ ,$$
$$R^1_{322} = 0 \ ,$$
$$R^1_{323} = 0 \ ,$$
$$R^1_{330} = 0 \ ,$$
$$R^1_{331} = 1.0r^2 \left(-k - \left(\frac{d}{dt}a(t)\right)^2\right) \sin^2(\theta) \ ,$$
$$R^1_{332} = 0 \ ,$$
$$R^1_{333} = 0 \ ,$$
$$R^2_{000} = 0 \ ,$$
$$R^2_{001} = 0 \ ,$$
$$R^2_{002} = \frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$
$$R^2_{003} = 0 \ ,$$
$$R^2_{010} = 0 \ ,$$
$$R^2_{011} = 0 \ ,$$
$$R^2_{012} = 0 \ ,$$
$$R^2_{013} = 0 \ ,$$

$$R^2_{020} = -\frac{1.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$
$$R^2_{021} = 0 \ ,$$
$$R^2_{022} = 0 \ ,$$
$$R^2_{023} = 0 \ ,$$
$$R^2_{030} = 0 \ ,$$
$$R^2_{031} = 0 \ ,$$
$$R^2_{032} = 0 \ ,$$
$$R^2_{033} = 0 \ ,$$
$$R^2_{100} = 0 \ ,$$
$$R^2_{101} = 0 \ ,$$
$$R^2_{102} = 0 \ ,$$
$$R^2_{103} = 0 \ ,$$
$$R^2_{110} = 0 \ ,$$
$$R^2_{111} = 0 \ ,$$
$$R^2_{112} = \frac{1.0\left(k + \left(\frac{d}{dt}a(t)\right)^2\right)}{kr^2 - 1} \ ,$$
$$R^2_{113} = 0 \ ,$$
$$R^2_{120} = 0 \ ,$$
$$R^2_{121} = \frac{1.0\left(-k - \left(\frac{d}{dt}a(t)\right)^2\right)}{kr^2 - 1} \ ,$$
$$R^2_{122} = 0 \ ,$$
$$R^2_{123} = 0 \ ,$$
$$R^2_{130} = 0 \ ,$$
$$R^2_{131} = 0 \ ,$$
$$R^2_{132} = 0 \ ,$$
$$R^2_{133} = 0 \ ,$$
$$R^2_{200} = 0 \ ,$$
$$R^2_{201} = 0 \ ,$$
$$R^2_{202} = 0 \ ,$$

$$R^2_{203} = 0 \ ,$$
$$R^2_{210} = 0 \ ,$$
$$R^2_{211} = 0 \ ,$$
$$R^2_{212} = 0 \ ,$$
$$R^2_{213} = 0 \ ,$$
$$R^2_{220} = 0 \ ,$$
$$R^2_{221} = 0 \ ,$$
$$R^2_{222} = 0 \ ,$$
$$R^2_{223} = 0 \ ,$$
$$R^2_{230} = 0 \ ,$$
$$R^2_{231} = 0 \ ,$$
$$R^2_{232} = 0 \ ,$$
$$R^2_{233} = 0 \ ,$$
$$R^2_{300} = 0 \ ,$$
$$R^2_{301} = 0 \ ,$$
$$R^2_{302} = 0 \ ,$$
$$R^2_{303} = 0 \ ,$$
$$R^2_{310} = 0 \ ,$$
$$R^2_{311} = 0 \ ,$$
$$R^2_{312} = 0 \ ,$$
$$R^2_{313} = 0 \ ,$$
$$R^2_{320} = 0 \ ,$$
$$R^2_{321} = 0 \ ,$$
$$R^2_{322} = 0 \ ,$$
$$R^2_{323} = 1.0r^2 \left(k + \left(\frac{d}{dt}a(t)\right)^2\right) \sin^2(\theta) \ ,$$
$$R^2_{330} = 0 \ ,$$
$$R^2_{331} = 0 \ ,$$
$$R^2_{332} = -1.0r^2 \left(k + \left(\frac{d}{dt}a(t)\right)^2\right) \sin^2(\theta) \ ,$$

$$R^2_{333} = 0 \ ,$$
$$R^3_{000} = 0 \ ,$$
$$R^3_{001} = 0 \ ,$$
$$R^3_{002} = 0 \ ,$$
$$R^3_{003} = \frac{1.0 \frac{d^2}{dt^2} a(t)}{a(t)} \ ,$$
$$R^3_{010} = 0 \ ,$$
$$R^3_{011} = 0 \ ,$$
$$R^3_{012} = 0 \ ,$$
$$R^3_{013} = 0 \ ,$$
$$R^3_{020} = 0 \ ,$$
$$R^3_{021} = 0 \ ,$$
$$R^3_{022} = 0 \ ,$$
$$R^3_{023} = 0 \ ,$$
$$R^3_{030} = -\frac{1.0 \frac{d^2}{dt^2} a(t)}{a(t)} \ ,$$
$$R^3_{031} = 0 \ ,$$
$$R^3_{032} = 0 \ ,$$
$$R^3_{033} = 0 \ ,$$
$$R^3_{100} = 0 \ ,$$
$$R^3_{101} = 0 \ ,$$
$$R^3_{102} = 0 \ ,$$
$$R^3_{103} = 0 \ ,$$
$$R^3_{110} = 0 \ ,$$
$$R^3_{111} = 0 \ ,$$
$$R^3_{112} = 0 \ ,$$
$$R^3_{113} = \frac{1.0 \left(k + \left(\frac{d}{dt} a(t)\right)^2\right)}{kr^2 - 1} \ ,$$
$$R^3_{120} = 0 \ ,$$
$$R^3_{121} = 0 \ ,$$

$$R^3_{122} = 0 \ ,$$
$$R^3_{123} = 0 \ ,$$
$$R^3_{130} = 0 \ ,$$
$$R^3_{131} = \frac{1.0\left(-k - \left(\frac{d}{dt}a(t)\right)^2\right)}{kr^2 - 1} \ ,$$
$$R^3_{132} = 0 \ ,$$
$$R^3_{133} = 0 \ ,$$
$$R^3_{200} = 0 \ ,$$
$$R^3_{201} = 0 \ ,$$
$$R^3_{202} = 0 \ ,$$
$$R^3_{203} = 0 \ ,$$
$$R^3_{210} = 0 \ ,$$
$$R^3_{211} = 0 \ ,$$
$$R^3_{212} = 0 \ ,$$
$$R^3_{213} = 0 \ ,$$
$$R^3_{220} = 0 \ ,$$
$$R^3_{221} = 0 \ ,$$
$$R^3_{222} = 0 \ ,$$
$$R^3_{223} = 1.0r^2\left(-k - \left(\frac{d}{dt}a(t)\right)^2\right) \ ,$$
$$R^3_{230} = 0 \ ,$$
$$R^3_{231} = 0 \ ,$$
$$R^3_{232} = 1.0r^2\left(k + \left(\frac{d}{dt}a(t)\right)^2\right) \ ,$$
$$R^3_{233} = 0 \ ,$$
$$R^3_{300} = 0 \ ,$$
$$R^3_{301} = 0 \ ,$$
$$R^3_{302} = 0 \ ,$$
$$R^3_{303} = 0 \ ,$$
$$R^3_{310} = 0 \ ,$$

$$R^3_{311} = 0 \ ,$$
$$R^3_{312} = 0 \ ,$$
$$R^3_{313} = 0 \ ,$$
$$R^3_{320} = 0 \ ,$$
$$R^3_{321} = 0 \ ,$$
$$R^3_{322} = 0 \ ,$$
$$R^3_{323} = 0 \ ,$$
$$R^3_{330} = 0 \ ,$$
$$R^3_{331} = 0 \ ,$$
$$R^3_{332} = 0 \ ,$$
$$R^3_{333} = 0 \ ,$$
$$.$$

3.3 Ricci tensor

The Ricci tensor is

$$R_{00} = -\frac{3.0\frac{d^2}{dt^2}a(t)}{a(t)} \ ,$$
$$R_{01} = 0 \ ,$$
$$R_{02} = 0 \ ,$$
$$R_{03} = 0 \ ,$$
$$R_{10} = 0 \ ,$$
$$R_{11} = \frac{-2.0k - 1.0a(t)\frac{d^2}{dt^2}a(t) - 2.0\left(\frac{d}{dt}a(t)\right)^2}{kr^2 - 1} \ ,$$
$$R_{12} = 0 \ ,$$
$$R_{13} = 0 \ ,$$
$$R_{20} = 0 \ ,$$
$$R_{21} = 0 \ ,$$
$$R_{22} = r^2 \cdot \left(2.0k + 1.0a(t)\frac{d^2}{dt^2}a(t) + 2.0\left(\frac{d}{dt}a(t)\right)^2\right) \ ,$$
$$R_{23} = 0 \ ,$$
$$R_{30} = 0 \ ,$$

$$R_{31} = 0 \ ,$$

$$R_{32} = 0 \ ,$$

$$R_{33} = r^2 \cdot \left(2.0k + 1.0a(t)\frac{d^2}{dt^2}a(t) + 2.0\left(\frac{d}{dt}a(t)\right)^2 \right) \sin^2(\theta) \ ,$$

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3.4 Ricci scalar

The Ricci scalar is

$$R = \frac{6.0\left(k + a(t)\frac{d^2}{dt^2}a(t) + \left(\frac{d}{dt}a(t)\right)^2\right)}{a^2(t)} \ .$$

3.5 Einstein tensor

The Einstein tensor is

$$G_{00} = \frac{3.0\left(-k - 2a(t)\frac{d^2}{dt^2}a(t) - \left(\frac{d}{dt}a(t)\right)^2\right)}{a^2(t)} \ ,$$

$$G_{01} = 0 \ ,$$

$$G_{02} = 0 \ ,$$

$$G_{03} = 0 \ ,$$

$$G_{10} = 0 \ ,$$

$$G_{11} = \frac{-5.0k - 4.0a(t)\frac{d^2}{dt^2}a(t) - 5.0\left(\frac{d}{dt}a(t)\right)^2}{kr^2 - 1} \ ,$$

$$G_{12} = 0 \ ,$$

$$G_{13} = 0 \ ,$$

$$G_{20} = 0 \ ,$$

$$G_{21} = 0 \ ,$$

$$G_{22} = r^2 \cdot \left(5.0k + 4.0a(t)\frac{d^2}{dt^2}a(t) + 5.0\left(\frac{d}{dt}a(t)\right)^2 \right) \ ,$$

$$G_{23} = 0 \ ,$$

$$G_{30} = 0 \ ,$$

$$G_{31} = 0 \ ,$$

$$G_{32} = 0 \ ,$$

$$G_{33} = r^2 \cdot \left(5.0k + 4.0a(t)\frac{d^2}{dt^2}a(t) + 5.0\left(\frac{d}{dt}a(t)\right)^2 \right) \sin^2(\theta) \ ,$$

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